

**TECHNICAL & ECONOMIC WHITEPAPER**

Causeway Technical & Economic Whitepaper

A market intelligence layer for prediction markets: from market-data reasoning to source verification, swarm simulation, and user-governed decision workflows.

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This whitepaper describes Causeway's product thesis, technical architecture, roadmap, economic model, and governance boundary. It is not investment advice, legal advice, brokerage material, or a promise of return. Prediction markets are risky; any real action should remain user-directed and user-confirmed.

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01 / EXECUTIVE SUMMARY

Executive Summary

Causeway is a market intelligence layer for prediction markets. Its core thesis is that prediction markets are not isolated pages; they are networks of probability shaped by real-world events. A policy signal, sports result, company announcement, legal filing, or on-chain event rarely affects only one market. It casts a market shadow across related narratives, assets, and participant expectations.

Current prediction-market interfaces expose prices, titles, volume, and rules, but they do not provide the reasoning layer between an event and a reviewable decision. Users need structured inference: what may be affected, through which path, with what confidence, which assumptions remain unverified, and which actions should be previewed rather than executed immediately.

Causeway's role: Causeway is not an autonomous betting bot. It is a system for event reasoning, market mapping, source verification, simulation, previews, and user-governed decision workflows.

Version 1.1 expands the original technical-economic whitepaper with three major additions: the market-shadow thesis, a five-phase roadmap from market data to optional delegated execution, and the long-term vision of a swarm intelligence prediction engine that turns multi-agent simulations into professional prediction reports.

02 / MARKET THESIS

Market Problem and Product Thesis

2.1 The Interface Gap

Prediction-market prices aggregate expectations, but most interfaces remain market-by-market. For complex events, the real user problem is cross-market: did the event change related probabilities, has that change already been priced in, which markets lag, and which links are merely narrative noise?

2.2 Causeway's Principles

PRINCIPLE 01**Every event has a market shadow**

A single event can ripple through policy, macro, sports, crypto, election, and company narratives.

PRINCIPLE 02**Reasoning must be visible**

Useful AI should expose assumptions, confidence, uncertainty, and the reasoning path behind each market connection.

PRINCIPLE 03**Speed needs governance**

Fast reaction is valuable only when preview, confirmation, audit, and permission boundaries remain clear.

2.3 Target Outcome

Causeway organizes event sensing, market mapping, AI reasoning, source verification, action previews, and user confirmation into one workflow. The near-term objective is better prediction-market analysis and previews. The mid-term objective is a verifiable intelligence layer. The long-term objective is a swarm intelligence engine capable of simulating how the world may move.

03 / PRODUCT DEFINITION

Product Definition

Causeway starts with Polymarket market data and turns a user-selected thesis or event into a structured market response plan. The plan includes related markets, impact paths, reasoning, confidence, risk warnings, source state, preview limits, and audit records.

Layer	Function	User Value
Market Data	Normalize market structure, prices, liquidity, rules, and book state.	Understand the market before touching the world.
Market Shadow	Map an event into related markets and causal paths.	See cross-market motion instead of isolated pages.
Reasoning	Generate assumptions, confidence, direction, and review states.	Make AI outputs auditable and reviewable.
Source Verification	Trace claims to authoritative sources and detect conflicts.	Reduce false information and model hallucination risk.
Governed Preview	Convert scenarios into guarded previews and user-confirmed actions.	Preserve control while increasing speed.

Boundary: Causeway v1.x does not default to autonomous execution. AI may reason, simulate, score, and preview, but custody, confirmation, and final action remain user governed.

System Architecture

4.1 Six Layers

- 1. **Market Data Foundation:** a normalized, queryable, auditable market data base.
- 2. **Market Graph:** relationships based on categories, semantics, historical behavior, liquidity, and AI reasoning.
- 3. **Reasoning Engine:** multi-layer impact paths, scores, explanations, and review states.
- 4. **Source Object Layer:** normalized objects for news, filings, chain data, sports data, and official releases.
- 5. **Verification Layer:** authority-source checks, conflict detection, and freshness validation.
- 6. **Governed Preview & Execution:** previews, constraints, audit records, and user confirmation.

4.2 Core Objects

Object	Meaning	Key Fields
Market	A tradable or observable prediction-market unit.	marketId, question, category, rules, volume, liquidity, status.
Thesis	The event, hypothesis, or market idea the user wants to understand.	title, scope, timeWindow, riskMode, rootContext.
Market Shadow	The set of markets and paths affected by an event.	nodes, edges, impactDirection, confidence, reason.
Source Object	A traceable and verifiable information object.	origin, timestamp, entity, claim, rawPayload, confidence.
Preview	A reviewable plan before real action.	actionType, limits, checks, expiresAt, auditTrail.

AI Reasoning and Market Shadows

Causeway's AI layer should not merely produce an answer. It should produce a reviewable market-shadow graph that describes which markets are affected, how they are connected, why the link matters, how confident the system is, and what remains uncertain.

5.1 Scoring Dimensions

RELEVANCE Whether the event has a clear entity, theme, or economic relation to the market.	DIRECTION Whether the event supports, opposes, hedges, contradicts, or correlates with a market path.
CONFIDENCE The strength of evidence, comparability, and source quality.	TRADABILITY Whether the market is active, liquid, understandable, and previewable.

5.2 From Single Model to Agent Teams

Early versions may rely on a single model for structured outputs. Later versions should introduce agents with roles: analyst, skeptic, verifier, risk officer, and report writer. The result becomes a collective reasoning trace with visible consensus and conflict.

Source Objects and Verification

Speed is not useful if the source is wrong. Causeway's verification layer traces claims back to primary or authoritative sources: official releases, regulatory filings, sports-league data, court documents, government databases, company statements, and on-chain records.

Field	Purpose
origin	The original information source.
claim	A concrete factual claim to verify.
timestamp	Published, fetched, and last-updated time.
entities	People, organizations, markets, assets, teams, or regions.
confidence	A combined source-quality, freshness, and conflict score.
rawPayload	The original data retained for audit and replay.

Verification should affect workflow state. Low-confidence claims enter verify mode. Conflicting claims trigger warnings. High-impact but unverified claims may only enter dry-run preview. Expired sources require refresh.

07 / SWARM INTELLIGENCE

Swarm Intelligence Prediction Engine



Concept visual: multi-agent reasoning across simulated market worlds.

Causeway's long-term goal is not a faster trading button. It is a living intelligence layer for predicting how the world may move. Inspired by multi-agent simulation, the system can start from a seed event, build parallel market worlds, let agents with roles and memory interact, and turn consensus and conflict into a professional prediction report.

SIMULATE**Build parallel market worlds**

Model how events, narratives, incentives, and participants interact across environments.

EVOLVE**Let many agents reason together**

Specialized agents expose hidden paths, disagreements, and risks.

PREDICT**Turn simulation into decision intelligence**

Output possible futures, probability shifts, assumptions, source confidence, and review paths.

08 / ROADMAP

Five-Phase Roadmap

Phase 01

Now

Market Data Foundation

Phase one focuses on Polymarket market data first. External sources are a later source-object layer. The system must understand market structure, prices, liquidity, books, rules, and state before ingesting external noise.

Phase 02

Reasoning

Reasoning Model

Build a stronger model that maps one event into all relevant markets and scores relevance, direction, confidence, risk, and suggested review action.

Phase 03

Real-time

Real-Time Scenario Generation

Ingest real-time news streams and generate cross-market scripts: what happened, which themes matter, which markets are affected, and which confirmation signals are missing.

Phase 04

Verification

Source Verification Layer

Before action, trace claims to primary sources, detect conflicts, check freshness, and reduce hallucination and misinformation risk.

Phase 05

Future

Delegated AI Execution

Users may optionally delegate bounded authority to AI under explicit limits: market categories, order size, loss budget, data-source requirements, expiry, revocation, and audit.

Governance, Risk, and Control

- AI may reason, rank, simulate, and preview, but should not bypass user confirmation.
- Every real action requires a preview with path, risk checks, limits, and audit summary.
- The system must distinguish dry-run, guarded preview, user-confirmed action, and delegated execution.
- Delegation must be optional, bounded, revocable, expiring, and auditable.

Risk	Control
Information risk	Source Objects, authority-source verification, conflict detection.
Reasoning risk	Confidence thresholds, skeptic agents, review states, path explanations.
Market risk	Liquidity, slippage, market state, rules, and freshness checks.
Execution risk	User confirmation, permission boundaries, budgets, expiry, and revocation.
System risk	Structured validation, idempotency, audit logs, and capability fallback.

Economic Model

10.1 Target Users

- Advanced prediction-market users who need to understand event impact quickly.
- Researchers and strategy builders who need replayable reasoning and audit records.
- Small funds or trading teams requiring collaboration, limits, previews, and reporting.
- Developers who need market graph, source object, and simulation APIs.

10.2 Revenue Lines

Model	Description	Phase
Subscription	Market-shadow maps, inference runs, monitoring, and reports.	1-3
API / Data	Market graph, source verification, and prediction-report endpoints.	2-4
Team Workspace	Collaboration, audit, permissioning, and report export.	3-5
Strategy Infrastructure	Structured strategies with mandates, eligibility rules, risk budgets, and reporting.	4-5

11 / COMPLIANCE BOUNDARY

Data, Audit, and Compliance Boundary

Causeway should be designed as workflow software, a market intelligence system, and a user-governed decision tool. It should not be framed as a return-guaranteeing system or default autonomous trader.

- No AI output should be framed as investment advice.
- No prediction, strategy, or market outcome should be guaranteed.
- Simulation, preview, and real action must remain clearly distinct.
- Delegated execution requires explicit permission, scope, budget, expiry, and revocation.

Risks and Conclusion

Risk	Mitigation
False or stale information	Authority-source verification and source trails.
AI hallucination	Candidate-set constraints and structured output validation.
Insufficient liquidity	Fresh book checks and preview limits.
Over-automation	Human confirmation by default and bounded delegation.
Regulatory uncertainty	Clear disclaimers, configurable capabilities, and jurisdiction-aware deployment.

Causeway's value is not merely faster trading. It is a trustworthy intelligence infrastructure for prediction markets: market-data first, reasoning-visible, source-verifiable, simulation-aware, and user governed.

Long-term vision: Causeway evolves from a market reasoning tool into a swarm intelligence prediction engine. A user seeds an event; the system simulates possible worlds, organizes agent consensus and conflict, and outputs a professional prediction report.

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